Project Report: Reddit Clone (Ma3loma)

Submitted by:

- Basem Walid Talaat 23P0246 (Backend User auth)
- Amir Tamer Abdelrehim 23P0248 (FrontEnd communities)
- Mohamed Wael Badra 23P0059 (Backend communities)
- Omar Fouad Abdelhamid 23P0146 (Front end Posts)
- Omar Ahmed Abouraia 23P0100 (Front end user auth)
- Moaz Ahmed Mohamed Fathy 23P0049 (Backend posts)

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Subject: Full Stack Web Development Project Report

1. Introduction

This project is a fully functional clone of Reddit, a social news aggregation, web content rating, and discussion website. The platform allows users to submit content such as links, text posts, images, and videos, which are then voted up or down by other members.

The system is organized into communities (subreddits) covering a specific topic. Users can join these communities, interact via comments, and engage in real-time discussions. The project aims to replicate the core user experience of Reddit while implementing modern web development practices using the MERN stack.

2. Project Objectives

- To develop a robust **Content Management System** where users can create, edit, and delete posts.
- To implement a **Hierarchical Community Structure** allowing users to create and moderate their own communities.
- To build an interactive **Voting and Awarding System** (Karma) that determines content visibility.
- To enable **Real-time Communication** through a chat system and instant updates.
- To ensure **Media Handling** capabilities for image and video uploads using cloud storage.

3. Technology Stack

The project utilizes the **MERN** stack for a full Javascript development environment. **Frontend**

- **React.js (Vite):** For building a dynamic and responsive user interface.
- **Material UI (MUI):** For pre-built, accessible UI components (Buttons, Modals, Icons).
- **React Router DOM:** For client-side routing and navigation.
- **Axios:** For making HTTP requests to the backend API.
- **React Hot Toast:** For user notifications and error handling.

Backend

- **Node.js:** JavaScript runtime environment.
- **Express.js:** Web application framework for handling API routes and middleware.
- **Passport.js:** For authentication handling.

Database

- **MongoDB:** NoSQL database for storing unstructured data (users, posts, communities).

- **Mongoose:** ODM (Object Data Modeling) library for MongoDB validation and schema definition.

Third-Party Services

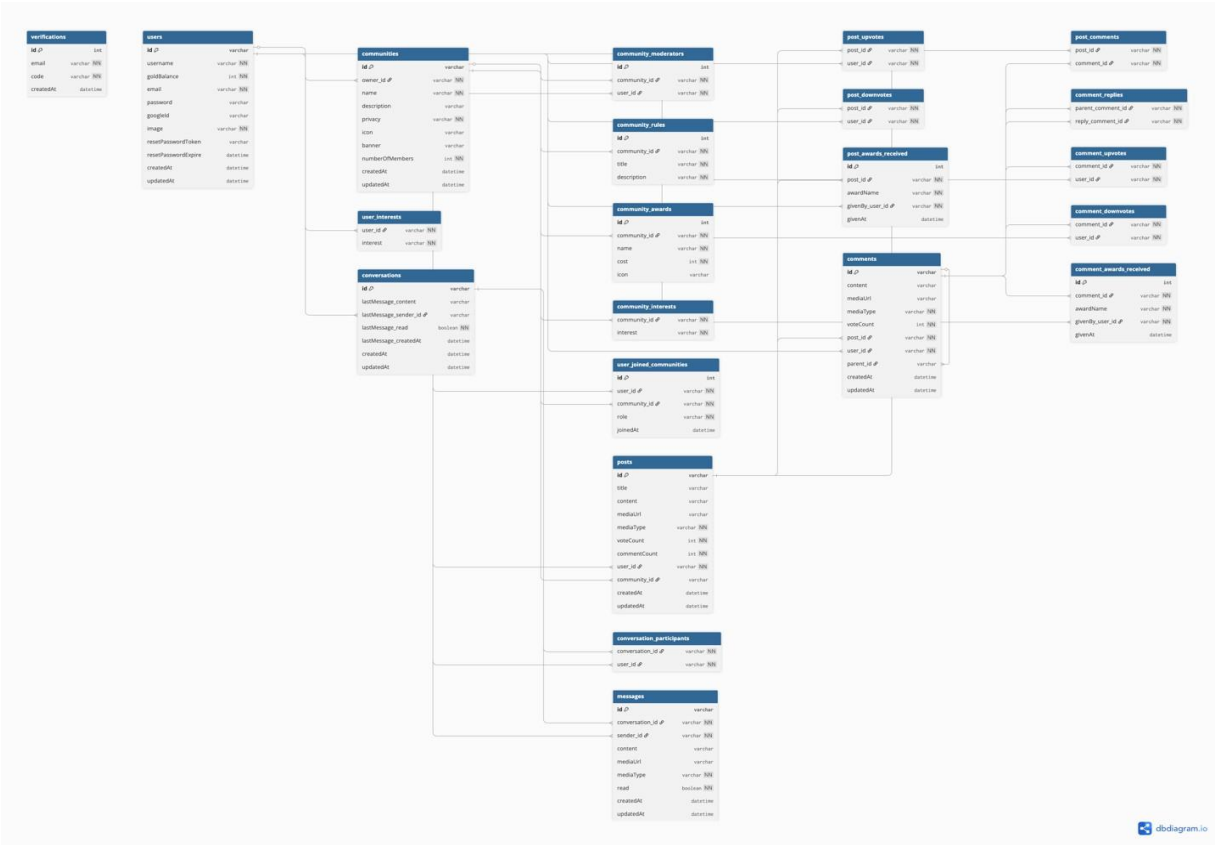
- **Cloudinary:** For secure cloud storage and optimization of user-uploaded images and videos.
- **Faker.js:** For generating realistic seed data for testing.

4. System Architecture & Design

4.1 Database Schema

The database is designed with relational references between documents to ensure data consistency.

- **Users:** Stores authentication details, profile images, bio, and referencing joined communities.
- **Communities:** Contains metadata (name, description, rules), arrays of moderators/members, and styling (icon, banner).
- **Posts:** Links to the User and Community. Stores content type (text/image/video), media URLs, upvotes/downvotes, and saved status.
- **Comments:** Supports nested structure (replies) by referencing parent comments.
- **Conversations/Messages:** Stores private chat history between users.



4.2 API Structure

The backend exposes RESTful endpoints organized by resource:

- `/api/auth`: Login, Signup, and Password Reset.
- `/api/posts`: CRUD operations for posts, voting, and awards.
- `/api/communities`: Community creation, joining, and fetching details.
- `/api/users`: Profile management and settings.
- `/api/chat`: Messaging functionality.

5. Key Features & Implementation

A. User Authentication & Profile

- **Secure Sign-up/Login:** Users can register accounts with secure password hashing.
- **Profile Customization:** Users can edit their avatars, banners, and bio descriptions.

B. Community Management

- **Creation Wizard:** A multi-step form allows users to create communities, set privacy levels (Public/Restricted/Private), and define topics.
- **Membership:** Users can join or leave communities, which affects their home feed content.

C. Advanced Post Creation

- **Rich Media Support:** The `CreatePost` component supports uploading images and videos directly to Cloudinary before submission, ensuring persistent media storage.
- **Editing:** Users can edit the text of their posts or update the media attachments after posting.

D. Interactions (Voting & Awards)

- **Voting System:** A dynamic upvote/downvote system updates the "vote count" in realtime, affecting post sorting.
- **Awards:** Users can gift awards (e.g., Gold, Silver, Fire) to posts or comments using a dedicated currency system. The awards are rendered visually on the content.

E. Commenting System

- **Nested Replies:** The system supports recursive comment threading, allowing for deep discussions similar to actual Reddit threads.

6. Challenges & Solutions

- **Media Persistence:**
  - *Challenge:* Initially, using `URL.createObjectURL` caused images to disappear after page refreshes because they were only stored in browser memory. *Solution:* Implemented a utility function `uploadToCloudinary` to upload files to the cloud immediately upon selection or submission, storing the permanent URL in MongoDB.
- **Data Seeding:**
  - *Challenge:* Testing the application required a large volume of consistent data (users, communities, posts). o*Solution:* Created a robust `seed.js` script using `Faker.js` to generate 100+ communities, 300+ users, and 5000+ posts with realistic relationships and voting history.

7. Future Enhancements

- **Dark Mode:** Implementing a system-wide theme toggle.
- **Moderation Tools:** Advanced dashboards for community moderators to ban users or remove content.
- **Notification System:** Real-time alerts for replies and mentions using Socket.io.

8. Conclusion

The "Ma3loma" Reddit Clone project successfully demonstrates a complex full-stack application. It integrates intricate database relationships with a polished frontend interface. The application handles real-world scenarios such as media uploads, authentication, and community management, satisfying all core functional requirements of a modern social media platform.

[Github Repository]